THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM ONE
041 BASIC MATHEMATICS

Time: 2½ hours **November 2024**

INSTRUCTIONS

1. This paper consists of ten (10) compulsory questions.
2. Show clearly all the working and answers in the space provided.
3. Four figure mathematical tables, geometric instruments and graph papers may be used where necessary.
4. Any unauthorized materials are not allowed in the examination room.
5. Write your examination number on every page of your answer sheets.

QUESTION #	MARKS	INITIALS OF EXAMINER
1		
2		
3		
4		
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7		
8		
9		
10		
Total		

1. (a) Write in words the least or the lowest sum of the square of 1 109 and the square root of 841

(b) Given 0.545454... at **Seypalm** secondary school represents boys. What fraction of students represents girls at that school?

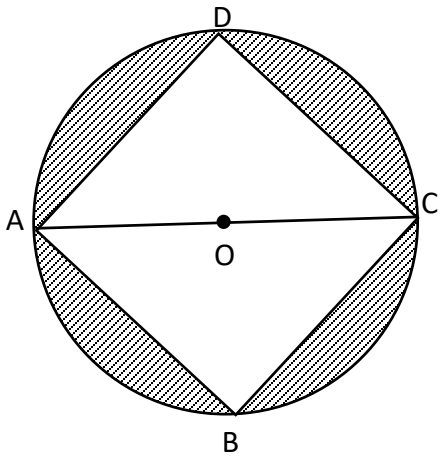
2. (a) Given 0.080579
   i. Write to 3 decimal places

   ii. Write to 3 significant figures

   (b) The length of of a wire measures 5 km 320 dam 409 cm. What is the length do 8 similar wires measure?

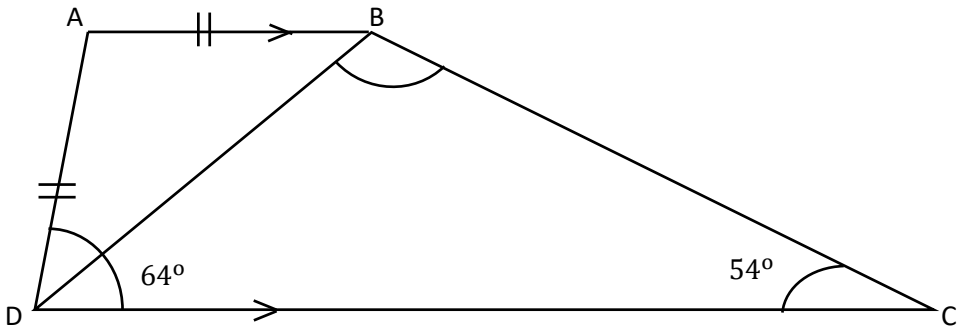
3. (a) In the following figure ,$\overline{AC}$ is a diameterof circle O with radius 5 cm. Find the area of the shaded region to the nearest whole number where $\overline{AD} = \overline{DC} = \overline{BC} = \overline{AB}$. Use $\pi = 3.14$

Figure 1



(b) In the following trapezium ABCD , $\overline{AB} \parallel \overline{DC}, \overline{AD} = \overline{AB}, \angle ADC = 64^{\circ}\ and\ \angle BCD = 54^{\circ}, find\ \angle DBC$

Figure 2



4. L(a) If $\left(\frac{9-r}{2}\right) = 14$, find the value of $2(r-9)$

(b) Solve the following simultaneous equations $\begin{cases} 15x - y = 9 \\ 7x - 2y = 12 \end{cases}$

5. (a) Share 80 000 shillings among **Seypalm**, Joseph, Juma, and Yasinta in the ratio of $2:4:5:9$

(b)Find the principal that would earn 108 000 shillings in 5 years at the rate of 5% per annum

6. (a) Given that: $2x - 3y + 7 = 0$
   i. Express the equation in form $y = mx + c$

   ii. Identify its gradient

   iii. Identify $y-$ intercept

(b) The equation of a line passes through points Q (, 2) and P(4 , 4). Calculate x if the gradient of the line is 2

7. (a) Seypalm carries a load of 4kg 30 g for one hour. How many hours does he need to carry the load of 20 kg 150 g?

(b) Graph the solution of the following: $-5 < 3x + 1 \leq 13$

8. (a) In a survey of 240 stusents, 37.5% prefer English, 0ne – sixth prefer Art, and the rest prefer Mathematics. How many more students prefer Mathematics than English does?

(b) Given: 21 , 31, 41, 51, 61, 71, 81, 91, 101, and 111

i. List prime numbers from the given numbers

ii. Using 1, 7, 13 and above numbers in 8b, write two equivalent fractions

9. (a)Decide which is longer between 40% of 3 days or 80% of 1128 minutes?

(b) Find the perimeter of a rectangle whose area is $50\ m^2$ and its width is 2.5 m

10. (a) i. Name a regular polygon whose sum of interior angles is 1080°

ii. Evaluate the following: $8 - 8 \times \frac{2\frac{1}{5} - 2\frac{1}{7}}{2 - \frac{1}{6 - \frac{1}{6}}}$

(b) Find the sum of the LCM and GCF of 18, 22, 15